Neel Gali

software engineer

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Professional Experience

Senior Software Engineer *Jul 2025 – Present*

Epik Solutions | Embedded Systems Optimization Platform

- Led the design and implementation of novel **compiler optimization techniques** for **MATLAB** and **Simulink** models, significantly improving **C++** code generation efficiency by 25% for **large-scale embedded systems**.
- Developed advanced **algorithmic skills** and **abstraction** layers within the **Embedded Coder** framework, achieving 30% reduction in memory footprint for generated code in automotive applications.
- Architected a modular code generation engine using modern **C++** and **object-oriented design principles**, enhancing extensibility and reducing development cycles by 15% for new target platforms.
- Implemented robust testing frameworks using **Google Test** for **Embedded Coder**'s generated code, ensuring high fidelity and reducing critical bugs in **large-scale embedded systems** by 40%.
- Optimized runtime performance of automatically generated **C++** code by integrating profile-guided optimization techniques, leading to a 10% speedup in critical aerospace control loop applications.
- Collaborated with customer support to analyze complex code generation issues for **MATLAB/Simulink** users, providing expert solutions and improving customer satisfaction scores by 18% quarterly.
- Designed and delivered **specifications** for new **Embedded Coder** features focusing on data type propagation and alias analysis, which accelerated **automatic code generation** by 12% in production.

Software Engineer *Jul 2022 – Jun 2025*

Samay Systems LLC | High-Performance Computing Solutions

- Implemented core modules for a high-performance simulation engine using modern **C++** and parallel processing paradigms, accelerating simulation runtimes by 35% for complex scientific models.
- Designed and developed new features for an embedded real-time operating system (RTOS) using **C++**, improving system responsiveness and reducing latency by 20% across critical applications.
- Optimized legacy **C++** codebases to adhere to stringent performance **requirements**, resulting in a 15% reduction in CPU utilization and enhanced stability for core system components.
- Contributed to the full **software development** lifecycle, including **writing specifications**, **coding**, and thorough **testing** for a mission-critical data acquisition system, achieving 99.9% uptime.
- Developed efficient data structures and **algorithmic skills** for processing large datasets in an HPC environment, reducing processing time for terabytes of sensor data by 25%.
- Integrated new hardware accelerators into the existing **C++** software stack, which boosted computational throughput by 50% for image processing tasks on specialized embedded devices.
- Collaborated with cross-functional teams to gather **requirements** for new software modules, delivering robust solutions that met tight deadlines and improved system integration by 18%.
- Authored comprehensive unit and integration tests using **Google Test** for critical **C++** components, improving code coverage by 20% and preventing major regressions in production.

Software Engineer *Jul 2020 – Aug 2022*

Zemoso Technologies | Enterprise Software Development

- Developed and maintained backend services using **Java** and **Spring Boot**, implementing **RESTful APIs** to support enterprise applications, handling over 10,000 requests per minute.
- Designed and implemented robust data processing modules in **C++** following **object-oriented design principles**, optimizing data ingestion pipelines by 20% for financial analytics platforms.
- Wrote comprehensive unit tests and integration tests using **JUnit** and **Mockito** for **Java** applications, increasing code quality and reducing production incidents by 25%.
- Collaborated on database schema design and query optimization for **PostgreSQL** to improve application performance, reducing report generation time by 15% for business intelligence tools.
- Participated in the full **software development** lifecycle, including gathering **requirements** and deploying features, ensuring timely delivery of critical functionalities for client-facing dashboards.
- Implemented efficient search algorithms and data structures in **C++** for a large-scale document management system, improving search query response times by 30% for users.

Skills

Languages: C++, Java
Cloud & Infra: PostgreSQL
Frameworks & Tools: MATLAB, Simulink, Embedded Coder, Google Test, Git, Spring Boot, JUnit, Mockito
Concepts: Compiler Optimization, Algorithmic Design, Object-Oriented Design (OOD), Abstraction, Code Generation, Embedded Systems, Software Development Lifecycle (SDLC), Performance Tuning, RESTful APIs

Education

Master's in Computer Science, Oklahoma State University, Stillwater
Bachelor's in Computer Science, JNTU Hyderabad, India